

Ashkenazi Population Structure Reassessed: Coordinate and Genotype-Level Reference-Panel Sensitivity in the Central Mediterranean

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Abstract

Population-genetic placement is conditional on the reference populations supplied to an analysis. This manuscript tests whether the familiar description of Ashkenazi Jews as intermediate between Europe and the Levant is stable when Italian Jewish, Southern Italian, Sicilian, Maltese, and Aegean comparators are included directly. The first empirical component is a linked suite of fifteen Global25 coordinate-space reference-panel experiments. These experiments include full modern competition panels, matched source omissions, reciprocal Italian Jewish targets, ten regional Ashkenazi targets, prehistoric controls, ancient Levant positive controls, and a full twenty-five-dimensional vector diagnostic. In the expanded modern Global25 panel, the source labeled Italki is the largest individual fitted source at 40.8, and the strict Italian Jewish and central Mediterranean block totals 54.6 at distance 0.00819655. Removing only Italki worsens distance by 12.49 percent and reallocates most of the missing coefficient to related Italian and central Mediterranean sources. Removing Italki together with Southern Italian, Sicilian, Maltese, and Aegean comparators worsens distance by 47.18 percent and inflates both Lombardy and Levantine-continuity proxies. The second empirical component uses AADR-derived individual-level genotype data rather than Global25 coordinates. It analyzes 369 primary-panel samples and 247 strict-pass samples across 18 population groups. FST is estimated on 184,385 shared post-QC autosomal SNPs; PCA and ADMIXTURE are run on 70,530 matched LD-pruned autosomal SNPs. Across PCA centroid distances, Hudson FST with chromosome jackknife uncertainty, replicated ADMIXTURE, and an equal-weight integrated rank synthesis, the closest or highest-ranked Ashkenazi comparators are concentrated among Sicilian, Turkish Jewish, Maltese, South Italian, Central Italian, Greek, and Cypriot populations. These results support a methodological conclusion: central Mediterranean comparators are necessary before Ashkenazi placement is interpreted as a simple Northern Italian-Levantine or Europe-Levant cline. The analyses are descriptive reference-panel sensitivity tests and are not converted into final historical source proportions.

Keywords: Ashkenazi Jews; population genetics; reference-panel sensitivity; comparator omission; central Mediterranean; Southern Italy; Italian Jews; Global25; AADR; PCA; FST; ADMIXTURE

1. Introduction

Genome-wide studies consistently identify Ashkenazi Jews as a genetically distinctive, endogamous population with ancestry related to both Europe and the eastern Mediterranean. Principal-component analyses, model-based clustering, identity-by-descent analyses, and ancient-DNA studies have repeatedly placed Ashkenazi samples near Southern European and Mediterranean populations while also detecting affinities with Levantine or other eastern Mediterranean groups. The central interpretive disagreement is not whether multiple ancestry-related signals exist, but how their geographic scale should be described and how much of the apparent Europe-Levant pattern depends on the reference populations selected for analysis.

The expression Europe-Levant intermediate is often treated as though it were an intrinsic property of the target population. In practice, intermediate placement is defined relative to a declared coordinate system and a chosen reference panel. A target can appear to lie between two poles when a more directly relevant population space is absent. This is particularly important in the Mediterranean, where Northern Italians, Tuscans, Sardinians, Southern

Italians, Sicilians, Maltese, Aegean Greeks, and Italian Jews are related but non-interchangeable comparators. If central Mediterranean variation is omitted, north-shifted Italian and Levantine references may become the nearest remaining directions and can bracket the missing space.

The comparator problem is visible at the beginning of modern Ashkenazi population genetics. Hammer et al. (2000) and Nebel et al. (2001) were foundational papers in the field and were widely interpreted as evidence for substantial shared paternal ancestry between Jewish and Middle Eastern populations. Behar et al. (2004), Shen et al. (2004), and Hammer et al. (2009) extended the Y-chromosomal literature with larger or more highly resolved datasets. Yet, with the important exception of Semino et al. (2004), these studies did not place Ashkenazi Jews beside non-Jewish Southern Italian regional samples capable of testing whether broadly Mediterranean E and J lineages were geographically specific to the Levant.

The limitation is consequential because haplogroups J1, J2, E-M78, and E-M123 are distributed across the eastern and central Mediterranean. Broad labels such as J or E-M35 cannot by themselves distinguish Levantine-specific ancestry from Southern Italian, Aegean, Anatolian, or wider Mediterranean structure. Semino et al. (2004) is important because it included Ashkenazi Jews together with regional samples from Apulia, Calabria, and Sicily at greater downstream resolution. In those data, Ashkenazi J2 and E-M123 frequencies fall within ranges reported for Southern Italian populations, while J1 remains secondary to J2. The overlap does not establish a unique Southern Italian paternal origin. It establishes that these lineages cannot be assigned a uniquely Levantine meaning by default when Southern European comparators are absent.

The same comparator architecture recurred when the field moved from uniparental markers to genome-wide autosomal data. The influential synthesis by Ostrer and Skorecki (2013) drew on Atzmon et al. (2010), Behar et al. (2010), Kopelman et al. (2009), Bray et al. (2010), Listman et al. (2010), and Campbell et al. (2012). These papers used different marker systems, sample sizes, and analytical methods, but the European side of the shared interpretive scaffold was repeatedly represented by Northern Italians, Tuscans, Sardinians, Basques, French, Russians, or other Northern and Central European populations. Non-Jewish Southern Italians and Sicilians were generally not represented as direct competing comparators.

The present manuscript therefore evaluates a reference-panel sensitivity thesis rather than a final demographic reconstruction. The primary hypothesis is that the apparent Ashkenazi Europe-Levant cline is sensitive to omission of Italian Jewish and central Mediterranean comparators. The directional prediction is not that every method will produce identical coefficients. It is that, when central Mediterranean comparators are restored, Ashkenazi placement will not be adequately summarized by a simple Northern Italian-Levantine bracket, and that removing those comparators will worsen fit or alter rank order in a predictable direction.

The analysis combines two data types without merging them at the input level. Global25 experiments use population-average coordinate data and source-panel perturbation. AADR genotype analysis uses individual-level SNP data, documented sample selection, QC, matched SNP construction, LD pruning, PCA, FST, ADMIXTURE, and rank synthesis. These approaches answer related but non-identical questions. Their convergence is interpreted as a robustness pattern; their outputs are not treated as interchangeable estimates of ancestry percentages.

Literature stage	Named studies	Reference-panel pattern	Interpretive consequence
Foundational Y-chromosomal studies	Hammer et al. (2000); Nebel et al. (2001); Behar et al. (2004); Shen et al. (2004); Hammer et al. (2009)	Jewish and Middle Eastern references were emphasized; non-Jewish Southern Italian regional comparators were generally absent.	Broad Mediterranean E and J lineages could be read as Levant-specific even though comparable Southern Italian variation was not directly tested.
Y-chromosomal exception	Semino et al. (2004)	Ashkenazi Jews were compared with Apulia, Calabria, Sicily, and related Mediterranean samples at greater downstream resolution.	Ashkenazi E-M123 and J2 frequencies overlapped Southern Italian regional values, weakening automatic Levant-only interpretation.

Literature stage	Named studies	Reference-panel pattern	Interpretive consequence
Foundational autosomal studies	Kopelman et al. (2009); Atzmon et al. (2010); Behar et al. (2010); Bray et al. (2010); Listman et al. (2010); Campbell et al. (2012)	Northern Italians, Tuscans, Sardinians, Basques, French, Russians, or North/Central Europeans were repeatedly available; non-Jewish Southern Italians and Sicilians were generally missing as direct competitors.	A repeated north-shifted European scaffold could make Ashkenazi samples appear more stably intermediate between Northern Italy/Europe and the Levant.
Modern Mediterranean-inclusive context	Tian et al. (2009); Zoossman-Diskin (2010); Xue et al. (2017); Drineas et al. (2019)	Italians, Greeks, Sicilians, Cretans, and related Mediterranean populations appear as close or informative comparators.	External results motivate, but do not replace, controlled panel restoration and omission tests.

Table 1. Recurring reference-panel design across foundational and contextual Ashkenazi population-genetic studies.

Test	Panel change or control	Directional prediction
Full modern competition	Italian Jewish, Southern Italian, Sicilian, Maltese, Aegean, Lombardy, Levantine, and Northern/Eastern European sources compete.	The largest fitted direction should fall within the Italian Jewish and central Mediterranean block if that space is directly relevant.
Italki omission	Remove only the Italki source while retaining related Mediterranean sources.	Most displaced weight should transfer to related Mediterranean sources, with a moderate increase in distance.
Wider omission	Remove Italki, Southern Italian, Sicilian, Maltese, and Aegean sources.	Lombardy and Levantine proxies should absorb omitted signal, and distance should worsen more strongly.
Reciprocal comparator	Model Italki as the target while excluding Ashkenazi sources.	Central Mediterranean sources should remain sufficient without later Northern/Eastern European host sources.
Regional replication	Repeat the expanded panel for ten regional Ashkenazi averages.	The category-level pattern should persist despite subgroup and sample-size differences.
Positive controls	Model ancient Levantine targets with appropriate earlier Levantine sources.	The framework should recover strong local Levantine continuity rather than forcing central Mediterranean sources into every target.

Table 2. Pre-specified tests and directional predictions for the reference-panel sensitivity analysis.

2. Materials and Methods

2.1 Study Design and Evidence Streams

The study is a secondary analysis of previously published or publicly distributed data. It introduces no new participant recruitment, biological sampling, genotyping, or sequencing. The Global25 component is a linked set of fifteen controlled coordinate-space experiments designed to test reference-panel sensitivity. The AADR component is an individual-level genotype analysis designed to test whether the same regional ordering appears when the evidence is expressed through SNP-level PCA, genome-wide FST, replicated ADMIXTURE, and cross-method rank synthesis.

Evidence stream	Input data	Analytical role
Global25 coordinate experiments	Population-average scaled Global25 coordinates and declared calculator source panels	Matched restoration/omission, reciprocal Italki, regional Ashkenazi, ancient controls, vector diagnostic
AA DR genotype analysis	Individual-level AADR-derived genotype panels after QC and SNP matching	PCA centroid distance, Hudson FST, ADMIXTURE, primary-versus-strict sensitivity, integrated comparator ranking
Formal source modeling boundary	Not performed in this manuscript	qpWave/qpAdm require separate source/right-population design and are not inferred from descriptive outputs

Table 3. Organization of the manuscript evidence. The two input data types are analyzed separately and compared only at the interpretive level.

2.2 Global25 Coordinate-Space Experiments

Global25 experiments used scaled population-average coordinates distributed through declared source files. The coordinate rows were used as distributed. The study did not generate or alter upstream genotype calls for the populations represented by Global25 coordinates, did not impute missing genotypes, and did not have access to individual-level genotype files for those coordinate rows. Marker-level quality filtering, sample exclusion rules, phasing, linkage-disequilibrium treatment, and construction of the underlying principal-component space were therefore inherited from the upstream coordinate-generation workflow. This limitation is one reason the Global25 analysis is framed as a coordinate-space sensitivity analysis rather than a raw-genotype demographic reconstruction.

Runs were recorded from Thierry Kerneur's ExploreYourDNA Atlantic Bronze Age calculator with cycles set to 0.25X, distance column off, printed zeroes off, fast mode off, and aggregate mode on. Displayed coefficients are nonnegative, constrained, source-panel-dependent similarity weights. Correlated sources can substitute for one another, and an unprinted or zero coefficient does not demonstrate historical absence. A modern proxy label is geographic shorthand for sampled coordinate variation and is not evidence that the named present-day population was a literal ancestral source. For matched omission experiments, fit deterioration was calculated as $100 \times (d_{\text{reduced}} - d_{\text{full}}) / d_{\text{full}}$, where d is the displayed residual distance.

The Global25 experiments were organized as connected controls. Experiments 1 and 8 used fixed prehistoric panels. Experiments 2 and 10 used broad modern competition panels. Experiments 3, 5, 11, and 12 removed declared source categories. Experiments 4, 6, and 13 provided direct-distance and reciprocal Italki comparisons. Experiments 7 and 14 evaluated ten regional Ashkenazi population averages. Experiments 9 and 15 tested ancient Levantine targets. Prehistoric and modern sources were not mixed indiscriminately within one interpretive panel.

Experiments	Role
1 and 8	Fixed prehistoric controls used to test behavior of deep ancestral components.
2 and 10	Full modern competition panels.
3, 5, 11, and 12	Matched omission tests and proxy-inflation diagnostics.
4, 6, and 13	Direct-distance and reciprocal Italki comparator tests.
7 and 14	Regional replication across ten Ashkenazi population averages.
9 and 15	Ancient Levant positive controls.

Table 4. Organization of the fifteen Global25 reference-panel sensitivity experiments.

2.3 AADR Individual-Level Genotype Analysis

The AADR genotype analysis used a deliberately constructed modern reference panel centered on Ashkenazi, central Mediterranean, Jewish comparator, Levantine, and broader European/Mediterranean populations. The primary panel retained the broad intended sample set after baseline quality control and contained 369 samples across 18 population groups. The strict-pass panel applied more restrictive sample filtering and contained 247 samples across the same population-label framework. The Ashkenazi target group contained seven samples in both panels; genotype results are therefore interpreted as reference-panel sensitivity evidence rather than as high-precision estimates of Ashkenazi population-wide parameters.

Genome-wide F_{ST} analyses used 184,385 shared post-QC autosomal SNPs. PCA and ADMIXTURE analyses used a matched LD-pruned autosomal subset of 70,530 SNPs. PCA was run on the matched LD-pruned genotype datasets. Population centroids were calculated in PC space, and Ashkenazi-to-comparator centroid distances were summarized across PC1-PC10. Primary-versus-strict stability was evaluated using Procrustes alignment across PC1-PC10.

Pairwise population differentiation was estimated on the shared post-QC autosomal SNP set. Hudson F_{ST} was calculated as a ratio of summed finite-sample-corrected numerators to summed denominators. Uncertainty was estimated by delete-one-autosome jackknife across chromosomes 1-22. Ashkenazi-versus-comparator rankings were cross-validated with PLINK Weir-Cockerham weighted F_{ST} .

Unsupervised ADMIXTURE was run independently in each panel for K=2 through K=12, with five deterministic random-seed replicates per K, ten-fold cross-validation, and eight computational threads. Replicates were aligned before averaging. Because ADMIXTURE components are model-based latent clusters, component labels were not interpreted as literal ancient populations or direct ancestry proportions.

For cross-method synthesis, Ashkenazi comparator ranks were integrated across PCA centroid distance, Hudson FST, and ADMIXTURE population-mean Q-profile distance. The three K values with the lowest mean cross-validation error in each panel were used for ADMIXTURE rank synthesis. Ranks were normalized within each method and panel, then averaged with cross-method and cross-panel dispersion retained as a stability measure.

Stage	Function	Manuscript role
01	Source-data acquisition, conversion, normalization, and organization.	Data provenance and reproducibility.
02	Sample identification, label harmonization, panel design, primary and strict-pass manifests.	Panel definition.
03	Sample and variant QC, exclusions, retained/rejected records, integrity checks.	Quality-control evidence.
04	Matched genotype panels, shared SNP sets, allele-order validation, LD-pruned datasets.	Analysis-ready dataset construction.
05	PCA coordinates, population centroids, Ashkenazi centroid distances, Procrustes stability.	Main genotype result.
06	Hudson FST, chromosome jackknife uncertainty, PLINK Weir-Cockerham validation.	Main genotype result.
07	ADMIXTURE K=2-12, five replicates per K per panel, 10-fold CV, aligned consensus.	Main genotype result.
08	Cross-method integrated comparator ranking and primary-versus-strict synthesis.	Main genotype result.

Table 5. AADR genotype-analysis stage map. Stages 01-04 establish provenance and analysis-ready data; Stages 05-08 produce PCA, FST, ADMIXTURE, and integrated-rank results.

Quantity	Value
Primary panel	369 samples
Strict-pass panel	247 samples
Population groups	18
FST SNP set	184,385 shared post-QC autosomal SNPs
PCA/ADMIXTURE SNP set	70,530 matched LD-pruned autosomal SNPs
ADMIXTURE design	K=2-12, five replicates per K per panel, 110 total runs, 10-fold CV

Table 6. AADR genotype panel dimensions and analytical design.

3. Results

3.1 Global25 Source-Panel Sensitivity

Experiment 10 provides the main full-panel Global25 comparison. At distance 0.00819655, the source labeled Italki is the largest individual fitted source at 40.8. The strict Italian Jewish and central Mediterranean block totals 54.6, with Lombardy at 9.8, Levantine continuity at 22.8, and Northern/Eastern Europe at 12.8. These values establish the baseline response when all declared source categories are allowed to compete; they are not treated as ancestry percentages.

Experiment 11 removes only Italki. Distance increases to 0.00922012, a deterioration of 12.49 percent. The strict central Mediterranean block remains 43.6 because related Mediterranean sources absorb much of the displaced coefficient. Lombardy rises to 14.4, Levantine continuity rises to 32.6, and Northern/Eastern Europe declines to 9.4. The result is consistent with source substitution among correlated Mediterranean populations rather than with a unique, literal contribution from one modern group.

Experiment 12 removes Italki together with Southern Italian, Sicilian, Maltese, and Aegean sources. Distance rises to 0.01206331, a deterioration of 47.18 percent relative to the full panel. The strict central Mediterranean block is no longer available; Lombardy rises from 9.8 to 38.0 and Levantine-continuity proxies rise from 22.8 to 52.0. The reduced panel therefore produces a more visually clinal Lombardy-Levantine solution while fitting the target substantially worse.

Experiment	Panel	Distance	Fit change
10	Full modern competition with Italki and central Mediterranean sources	0.00819655	Baseline
11	Same panel with Italki omitted	0.00922012	+12.49% vs. Exp. 10
12	Italki, Southern Italian, Sicilian, Maltese, and Aegean sources omitted	0.01206331	+47.18% vs. Exp. 10

Table 7. Global25 Experiments 10-12 matched panel perturbations. Fit change is relative to the full modern panel where available in the source table.

Global25 matched source-panel perturbation

Displayed coefficients are constrained coordinate-space similarity weights, not ancestry percentages.

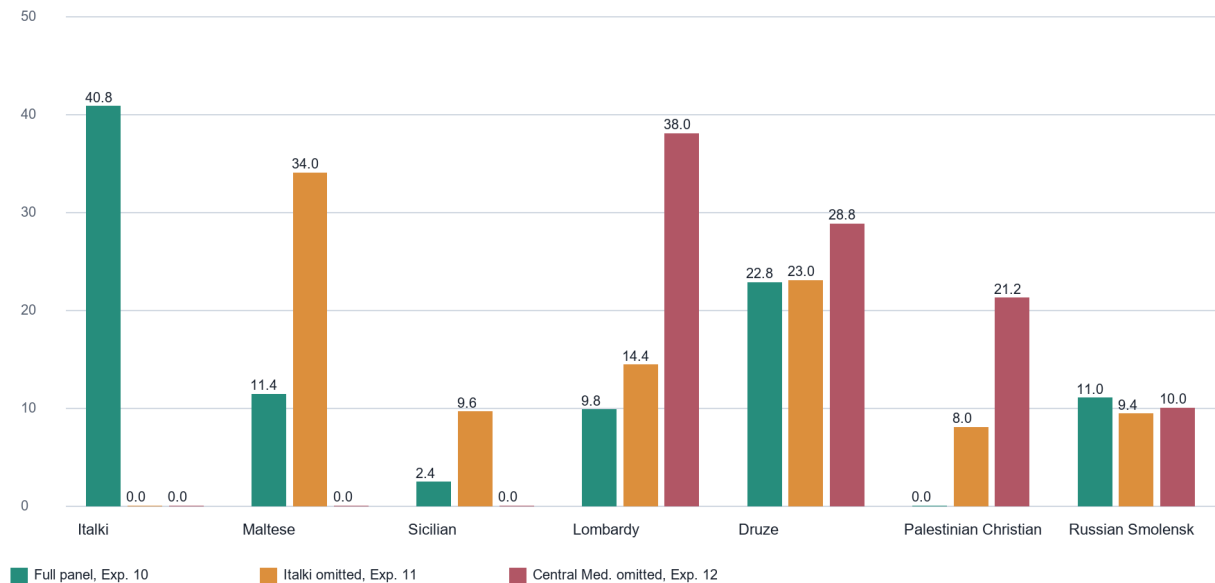


Figure 1. Global25 Experiments 10-12. Removing the central Mediterranean comparator space shifts fitted weight into Lombardy and Levantine-continuity proxies while worsening residual distance.

3.2 Reciprocal, Regional, and Ancient-Control Results

Reciprocal Italki models test whether the Italian-associated Ashkenazi placement is generated mainly by later Northern or Eastern European host populations. Experiment 6 models Italki without using Ashkenazi Jews as a source; the displayed solution resolves through Maltese, Rhodian Greek, eastern Sicilian, and Reggio Calabria sources. Experiment 13 expands the reciprocal panel by adding modern Levantine-continuity controls while continuing to exclude Ashkenazi sources. At distance 0.00984550, the displayed Italki solution contains 70.6 strict central Mediterranean, 4.2 Lombardy, and 25.2 Levantine continuity, with no Northern/Eastern European coefficient retained.

Experiments 7 and 14 apply the modern panel to ten named regional Ashkenazi population averages. In Experiment 14, the strict central Mediterranean block exceeds 50 in every region and ranges from 50.8 in Latvia to 76.4 in France, with an unweighted average of 58.6. The broader Italian and central Mediterranean block averages 62.5. The regional distributions constrain a simple host-population explanation because the category-level central Mediterranean

response persists despite subgroup and sample-size differences.

Ancient Levant positive controls show that the coordinate framework does not automatically force central Mediterranean sources into every target. Experiment 15 models three Iron Age coastal southern-Levantine averages against pre-Iron-Age Levantine site-period averages plus Mycenaean and central Mediterranean comparators. The unweighted target average is 89.8 pre-Iron-Age southern Levant, 7.3 Mycenaean composite, and 2.9 strict central Mediterranean. This result demonstrates that the framework recovers strong local Levantine continuity when appropriate earlier Levantine references are supplied.

Global25 Experiment 14 regional Ashkenazi stability

Strict central Mediterranean block exceeds 50 in all ten regional targets; unweighted average is 58.6.

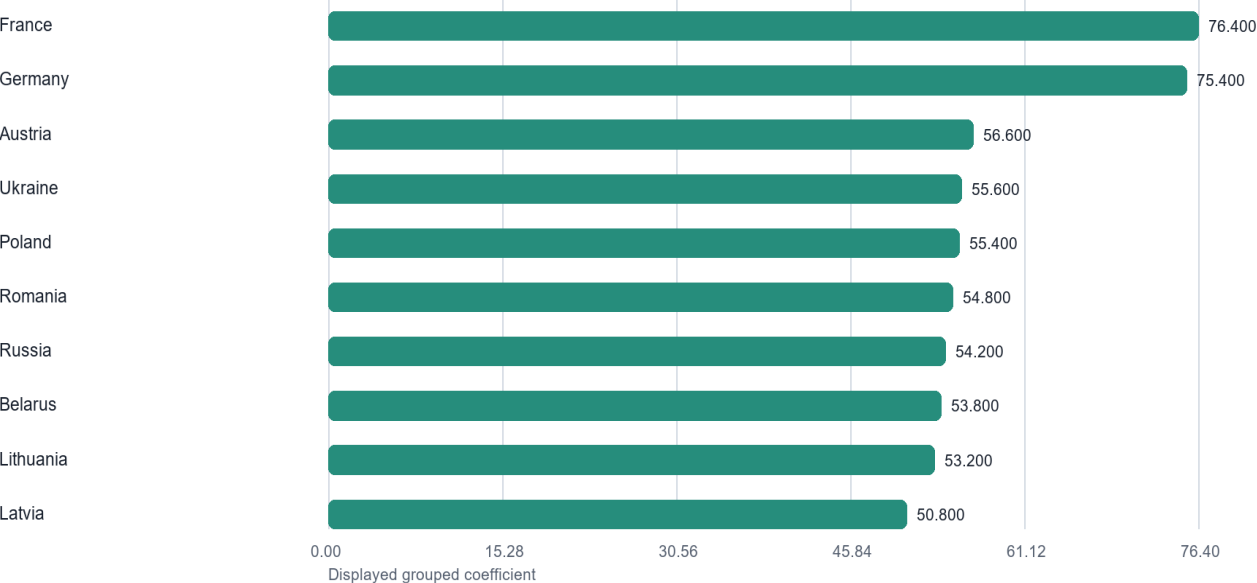


Figure 2. Global25 Experiment 14 regional Ashkenazi stability. The strict central Mediterranean block is retained above 50 in all ten regional Ashkenazi targets.

3.3 Convergence With Independent Published Evidence

The panel-sensitivity results are consistent with the recurring regional ordering in independent studies. This convergence is not statistical independence in a strict sense because several studies share overlapping reference-panel traditions, but it shows that Italian, Greek, Sicilian, Turkish Jewish, and broader central/eastern Mediterranean comparators repeatedly appear as relevant before the present controlled omission tests are considered.

Published genetic-distance analyses place Ashkenazi Jews closer to Italian and Greek populations than to sampled Levantine populations in some frameworks. Identity-by-descent analyses distinguish older shared structure from more recent gene flow, constraining simple interpretations based solely on geometric proximity. Ancient-DNA studies show that major Ashkenazi-associated ancestry components and founder effects were already present in medieval communities. These results motivate the present sensitivity analysis; they are not combined into a meta-analytic ancestry proportion.

Study	Method/data	Reported regional ordering or model	Interpretive constraint
Tian et al. (2009)	Autosomal FST among European and Middle Eastern populations.	Ashkenazi Jews show lower distance to Italians or Greeks than to sampled Palestinians.	Mediterranean proximity is not unique to this manuscript's coordinate experiment.
Zoossman-Diskin (2010)	Independent autosomal distance framework.	Ashkenazi Jews are reported closer to Italians/Greeks than to sampled Levantines in that framework.	Supports the need to test Southern European comparators directly.

Study	Method/data	Reported regional ordering or model	Interpretive constraint
Xue et al. (2017)	Identity-by-descent segment lengths and ancestry timing.	European-related mixture is constrained by segment length and time depth rather than PCA position alone.	Geometric proximity should not be overread as one simple recent source.
Lerga-Jaso et al. (2025)	Orchestra local-ancestry inference.	Broad Italian assignment is reported as the largest component, with smaller Levantine and other categories.	Local-ancestry categories are model-dependent but independently motivate Italian/Mediterranean comparator testing.
Waldman et al. (2022)	Medieval Erfurt ancient DNA and published qpAdm models.	Plausible South Italian models average approximately 65 percent South Italian, 19 percent Middle Eastern, and 16 percent Eastern European, with model variation.	Major ancestry components pre-date the fourteenth century and should be tested with ancient context.
Brace et al. (2022)	Medieval Norwich ancient DNA and a proxy model.	One model reports approximately 67 percent Sicilian proxy and 33 percent Turkish Jewish proxy.	A small proxy-sensitive model supports central/eastern Mediterranean-associated components before the twelfth century.

Table 8. Independent published results that provide external context for the coordinate-space and genotype-panel sensitivity analyses. Reported values are treated as model- or panel-dependent constraints.

3.4 AADR PCA, FST, ADMIXTURE, and Integrated Synthesis

The AADR genotype analysis re-expresses the reference-panel question using individual-level SNP data rather than Global25 population averages. The primary panel contains 369 samples; the strict-pass panel contains 247 samples. The shared FST SNP set contains 184,385 post-QC autosomal SNPs; the matched LD-pruned SNP set for PCA and ADMIXTURE contains 70,530 autosomal SNPs. These dimensions make the AADR analysis an independent data representation, not a reformatting of the Global25 coordinate outputs.

In PCA, the nearest Ashkenazi population centroids using PC1-PC10 are concentrated among central and eastern Mediterranean comparators. In the primary panel, the five nearest centroids are Maltese, Sicilian, South Italian, Turkish Jewish, and Cypriot. In the strict-pass panel, the five nearest centroids are Maltese, Sicilian, Cypriot, Turkish Jewish, and Greek. The primary-versus-strict Procrustes RMSD across PC1-PC10 is 0.0543532167, indicating that the principal coordinate configuration changes under stricter filtering but retains the same broad comparator structure.

Panel	Top five Ashkenazi centroid comparators
Primary	Maltese, Sicilian, South Italian, Turkish Jewish, Cypriot
Strict-pass	Maltese, Sicilian, Cypriot, Turkish Jewish, Greek

Table 9. PCA centroid-distance ordering across primary and strict-pass AADR genotype panels.

AADR genotype analysis: PCA centroid distance

Primary-panel Ashkenazi-to-comparator centroid distances across PC1-PC10.

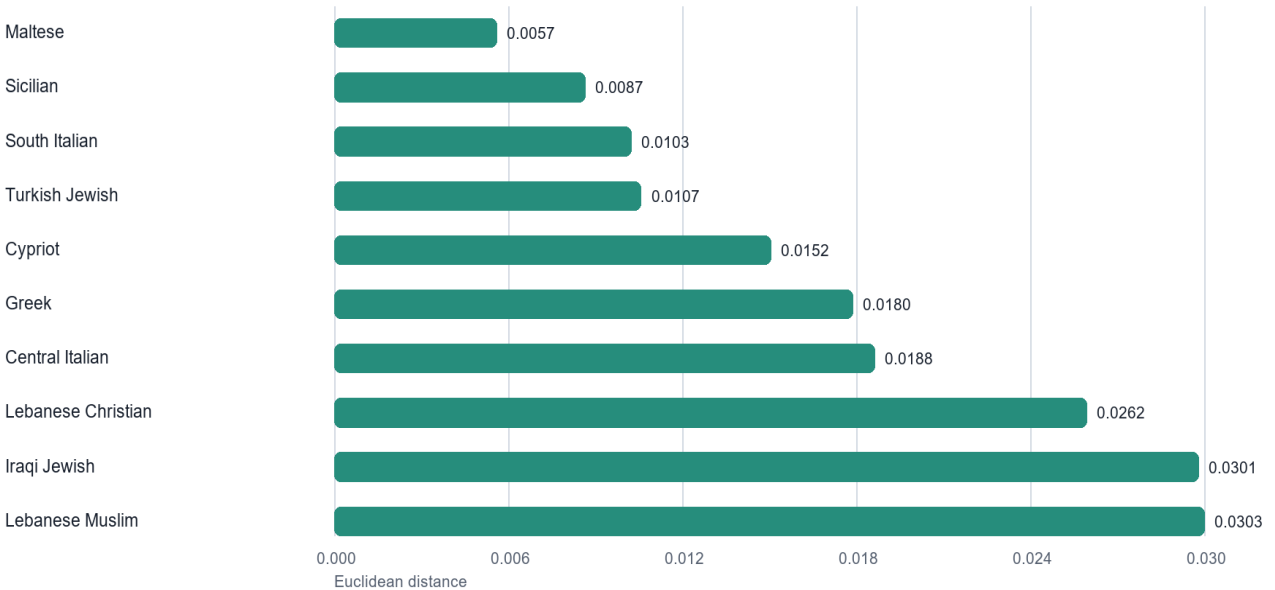


Figure 3. AADR primary-panel PCA centroid distances to Ashkenazi across PC1-PC10. Lower values indicate closer centroid placement within the tested genotype panel.

Genome-wide Hudson FST produces a closely related ordering. In both the primary and strict-pass panels, the five lowest Ashkenazi pairwise Hudson FST comparisons are Turkish Jewish, Sicilian, Central Italian, South Italian, and Greek. Across the 17 Ashkenazi-versus-comparator pairs, primary and strict Hudson FST rankings have Spearman $\rho=0.9093$. Hudson rankings are concordant with PLINK Weir-Cockerham weighted FST, with Spearman $\rho=0.9583$ in the primary panel and $\rho=0.9093$ in the strict-pass panel.

ADMIXTURE places Ashkenazi population-mean Q profiles nearest to central and eastern Mediterranean comparators, although the CV-supported K values differ between panels. In the primary panel, the lowest mean cross-validation error occurs at K=12, with K=12, K=11, and K=10 used as the supported K range for rank synthesis. In the strict-pass panel, the lowest mean cross-validation error occurs at K=2, with K=2, K=3, and K=4 used as the supported K range. This difference is treated as a sensitivity result rather than forced into one universal K.

Panel	Lowest Ashkenazi Hudson FST comparators
Primary	Turkish Jewish, Sicilian, Central Italian, South Italian, Greek
Strict-pass	Turkish Jewish, Sicilian, Central Italian, South Italian, Greek

Table 10. Hudson FST ordering across primary and strict-pass AADR genotype panels.

AADR genotype analysis: Hudson FST

Primary-panel Ashkenazi pairwise Hudson FST. Lower values indicate lower allele-frequency differentiation.

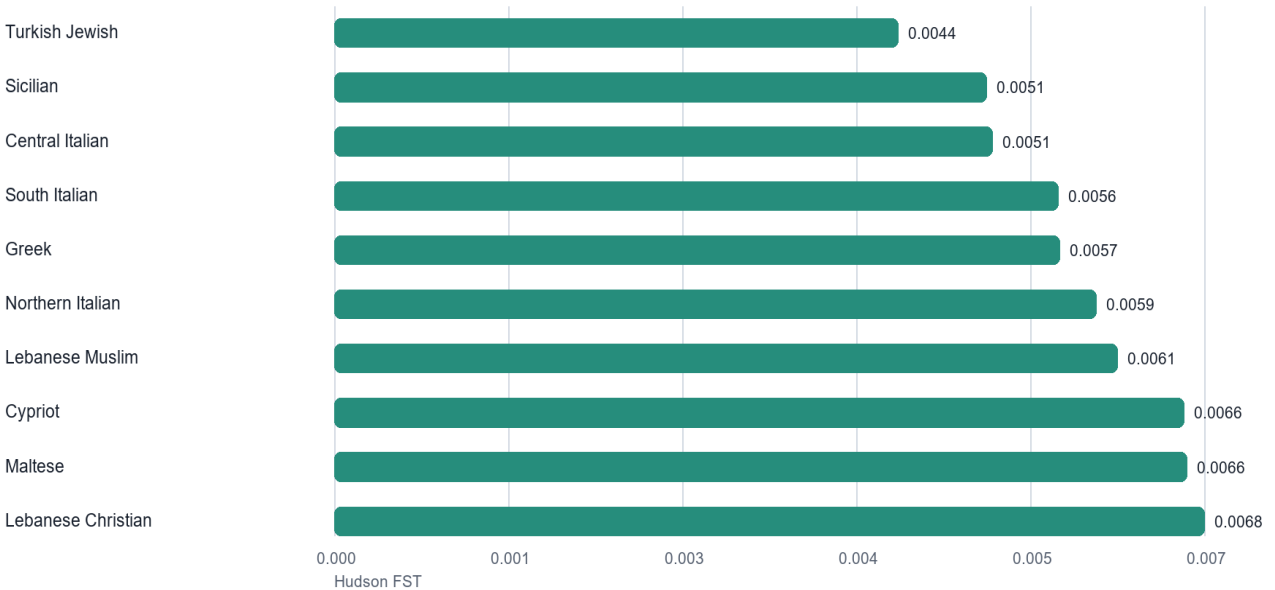


Figure 4. Primary-panel Ashkenazi pairwise Hudson FST for leading comparators. Lower FST indicates lower allele-frequency differentiation in the tested panel.

AADR genotype analysis: ADMIXTURE cross-validation

Primary panel minimum mean CV occurs at K=12; strict-pass minimum mean CV occurs at K=2.

Mean 10-fold CV error
0.565

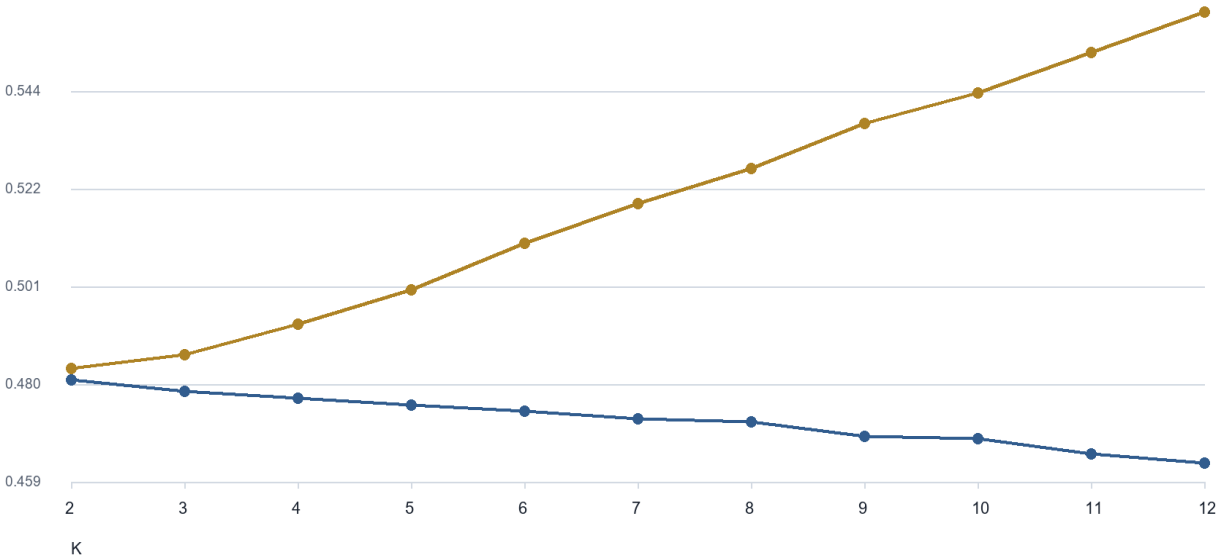


Figure 5. ADMIXTURE cross-validation across K=2-12. The different CV-supported K values are interpreted as panel sensitivity, not as one universal ancestry decomposition.

The equal-weight cross-method synthesis integrates PCA, FST, and ADMIXTURE ranks across the primary and strict-pass panels. The five highest-ranked comparators are Sicilian, Turkish Jewish, Maltese, South Italian, and Central Italian. Greek and Cypriot follow closely. Northern Italian, Tuscan, Bergamo Italian, and Levantine comparators are informative, but they rank lower or show greater panel sensitivity within the tested design.

The Global25 and AADR results do not estimate the same quantity. Global25 coefficients summarize coordinate-space approximation under declared source panels; PCA describes covariance, FST measures allele-frequency differentiation, ADMIXTURE estimates latent profiles within chosen K and panel settings, and integrated rank synthesis summarizes descriptive proximity across methods. The substantive convergence is therefore not a claim of identical coefficients. It is the repeated requirement for Italian Jewish and central Mediterranean comparators before Ashkenazi placement is interpreted geographically.

Rank	Comparator	Category	Mean rank	Stability SD
1	Sicilian	Central/eastern Mediterranean	0.062	0.000
2	Turkish Jewish	Jewish comparator	0.125	0.088
3	Maltese	Central/eastern Mediterranean	0.177	0.251
4	South Italian	Central/eastern Mediterranean	0.250	0.225
5	Central Italian	Central/eastern Mediterranean	0.271	0.106
6	Greek	Central/eastern Mediterranean	0.292	0.047
7	Cypriot	Central/eastern Mediterranean	0.302	0.127
8	Northern Italian	Central/eastern Mediterranean	0.479	0.123
9	Lebanese Christian	Levantine	0.500	0.072
10	Lebanese Muslim	Levantine	0.531	0.129

Table 11. Integrated AADR genotype comparator ranking across PCA centroid distance, Hudson FST, and ADMIXTURE Q-profile distance.

AADR genotype analysis: integrated comparator ranking

Lower normalized rank indicates closer cross-method and cross-panel placement to Ashkenazi.

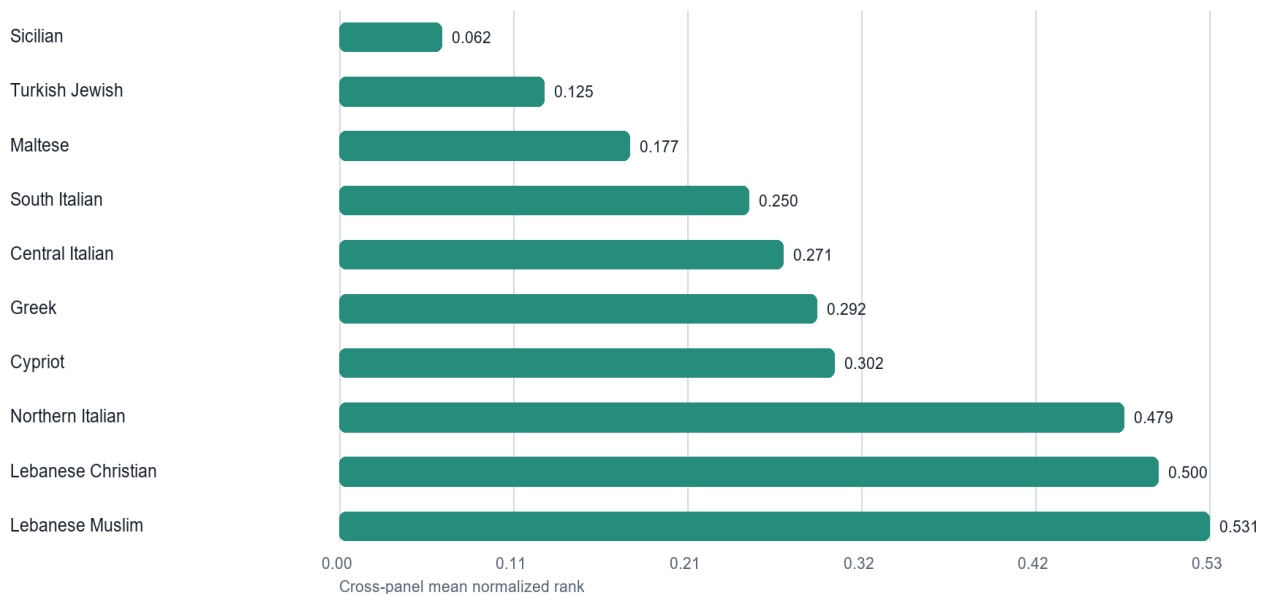


Figure 6. AADR integrated comparator ranking. Lower normalized rank indicates closer cross-method and cross-panel placement to Ashkenazi within the tested genotype design.

4. Discussion

The central result is methodological: the apparent Northern Italian-Levantine cline is not stable to the restoration of Italian Jewish and central Mediterranean comparators. In the full Global25 panel, the largest fitted direction is the

strict central Mediterranean block. Removing only Italki transfers most of its fitted weight to related Mediterranean sources and modestly worsens distance. Removing the wider central Mediterranean space produces a stronger Lombardy-Levantine combination but also a 47.18 percent deterioration in distance. The reduced model looks more clinal precisely because the source space nearest the target has been removed.

This behavior follows the geometry of constrained modeling. A target can be represented only by directions included in the panel. When the relevant central Mediterranean region is absent, a north-shifted Italian proxy and a Levantine proxy can bracket the missing space. The resulting coefficients may be visually persuasive while encoding a poorer approximation. Panel sensitivity should therefore be reported before proxy coefficients or intermediate placement are translated into historical narratives.

The AADR genotype analysis strengthens the reference-panel sensitivity argument because it uses individual-level genotype data and methods independent of the Global25 coordinate-fitting framework. PCA centroid distances, Hudson FST, and ADMIXTURE Q-profile distances do not estimate the same quantity, yet they converge on a similar regional ordering. Sicilian, Maltese, South Italian, Central Italian, Greek, Cypriot, and Turkish Jewish comparators remain central to Ashkenazi placement within the tested panels.

Italian Jews provide a reciprocal comparator for the narrow claim that the Italian-associated Ashkenazi component is primarily Northern Italian or is generated by later Northern/Eastern European host admixture. In reciprocal Global25 models, the Italki target retains a large central Mediterranean direction even when Ashkenazi sources are excluded and German, Polish, and Russian sources are permitted to compete. The shared placement therefore cannot be attributed solely to later Ashkenazi residence in the Rhineland or Eastern Europe.

Historical context makes the central Mediterranean hypothesis plausible but does not by itself assign a numerical genetic percentage. Jewish communities are documented in Southern Italy and Sicily from antiquity through the medieval period, including communities in Apulia, Calabria, Campania, Lucania, Syracuse, Palermo, Catania, Agrigento, and other centers. These communities participated in Greek- and Latin-speaking Mediterranean networks connected with the Aegean and Asia Minor. Italy was also a major staging area in the development of Jewish communities that later appear in northern France and the Rhineland.

The early Y-chromosomal literature should be read in this comparator context. Broad haplogroups span multiple Mediterranean regions, their frequencies can be altered by founder effects and endogamy, and low phylogenetic resolution can make widely distributed lineages appear more geographically diagnostic than the data support. Semino et al. (2004) is uniquely informative because it places Ashkenazi Jews beside multiple Southern Italian regional samples. Ashkenazi E-M123 and J2-M172 frequencies overlap Southern Italian regional values, and J2 exceeds J1 in Ashkenazi Jews and in every listed Southern Italian sample. This does not identify a unique source population, but it reduces the defensibility of assigning broadly Mediterranean paternal lineages to the Levant by default.

The correct inference is limited but consequential. These analyses do not erase eastern Mediterranean or Levant-related continuity. They change the geographic scale and certainty that can be assigned to proxy coefficients. A nonzero Levantine-continuity coefficient may combine genuine shared ancestry, source substitution, correction for drift, and compensation for the positions of other panel members. The most defensible statements concern comparative fit, redistribution under controlled omissions, rank ordering under genotype methods, and the target's location relative to the declared sources.

Formal qpWave and qpAdm analyses require a different design from the descriptive analyses presented here. Candidate sources and right populations must be declared in advance, outgroups must be justified, and models must be evaluated by feasibility, p-value, coefficient stability, standard errors, and nested-source competition. Present-day Jewish and Mediterranean populations may be useful comparators, but they should not be treated as literal ancient sources without appropriate ancient-DNA context. The descriptive evidence motivates formal testing but does not substitute for it.

Study	Reported Ashkenazi sample	Southern Italian comparator coverage	Main interpretive limitation
Hammer et al. (2000)	Ashkenazi sample included in Jewish/Middle Eastern Y-chromosomal comparison.	No direct Southern Italian regional comparator panel.	Broad Jewish-Middle Eastern sharing could not distinguish Levant-specific from wider Mediterranean lineages.
Nebel et al. (2001)	Small Ashkenazi sample in Y-chromosomal landscape study.	No direct Southern Italian regional comparator panel.	Limited sample size and comparator omission constrain geographic interpretation.
Behar et al. (2004)	Ashkenazi host-comparison design.	No direct Southern Italian regional comparator panel.	Broad haplogroup resolution limits Mediterranean substructure interpretation.
Shen et al. (2004)	Very small Ashkenazi sample in broader Israeli/Samaritan analysis.	No direct Southern Italian regional comparator panel.	Small sample and absent Southern Italian comparators limit inference.
Hammer et al. (2009)	Extended Jewish priesthood and Jewish lineage analysis.	Southern Italian regional samples not supplied for the relevant Ashkenazi comparisons.	Some summaries pool Ashkenazi and non-Ashkenazi Jewish samples.
Semino et al. (2004)	Ashkenazi Jews included with greater downstream resolution.	Apulia, Calabria, Cosenza Albanian community, Sicily.	The key exception: Southern Italian overlap can be tested directly.

Table 12. Foundational Y-chromosomal studies and Southern Italian comparator coverage. The table summarizes comparator availability rather than re-estimating paternal ancestry.

Population	Sample size	E-M78	E-M123	J1-M267	J2-M172
Ashkenazi Jews	77	7.8	11.7	3.9	23.2
Apulia	86	11.6	11.6	5.8	29.1
Calabria 1	57	22.8	1.8	8.8	22.8
Cosenza Albanian community	60	23.3	13.2	1.7	20.0
Sicily	54	11.1	5.6	3.7	16.7

Table 13. Semino et al. (2004) Y-chromosomal frequencies for Ashkenazi Jews and Southern Italian regional samples. Frequencies are percentages; regional overlap and heterogeneity do not prove a unique source population.

5. Limitations

Global25 is a compressed coordinate representation. The coordinate-space experiments cannot audit or repeat upstream marker-level quality-control, missingness filtering, phasing, or LD treatment from population-average coordinates alone. Their strength is the matched panel perturbation design, not formal parameter estimation.

Calculator outputs are constrained similarity weights without formal confidence intervals. Correlated sources can substitute for one another, and individual coefficients are not literal historical ancestry percentages. Several population averages have small or unequal sample sizes, including regional targets with $n=2$ and the Italki average with $n=9$; stability is therefore interpreted at the category level rather than as precision for any single regional coefficient.

The AADR genotype analysis reduces but does not eliminate proxy limitations. The panels are modern comparator panels, and modern populations are imperfect representatives of historical populations. Geographic labels identify sampled variation and do not establish direct descent from the named group. The Ashkenazi genotype target group contains seven samples in both the primary and strict-pass panels, so the genotype results are best read as reference-panel sensitivity evidence rather than as high-precision population parameter estimates.

PCA, FST, ADMIXTURE, Global25 fitting, and rank synthesis are descriptive tools with different mathematical targets. Their convergence supports the necessity of central Mediterranean comparators, but it does not establish admixture direction, date, source chronology, or exact ancestral proportions. No qpAdm or qpWave result is inferred

from ADMIXTURE components or from Global25 coefficients.

Historical and Y-chromosomal evidence provide context but cannot independently quantify genome-wide ancestry or identify a unique formation event. The manuscript therefore treats them as interpretive constraints around the autosomal and genotype-panel results, not as substitutes for genome-wide modeling.

6. Conclusions

The familiar Ashkenazi Europe-Levant cline is not stable under matched changes to the reference panel. With Italki, Southern Italian, Sicilian, Maltese, and Aegean sources present in the Global25 modern panel, they form the largest fitted direction. When those sources are removed, north-shifted Italian and Levantine proxies absorb the missing signal, and the fit becomes substantially worse.

The AADR individual-level genotype analysis supports the same methodological conclusion from a separate data representation. PCA centroid distances, Hudson FST, replicated ADMIXTURE, and integrated rank synthesis place the leading Ashkenazi comparators among Sicilian, Turkish Jewish, Maltese, South Italian, Central Italian, Greek, and Cypriot populations within the tested panels. This pattern does not make Global25 and AADR outputs interchangeable, but it does show that the central Mediterranean comparator requirement is not confined to one coordinate calculator.

The results do not eliminate Levant-related continuity. They require that such continuity be interpreted in a panel that also contains Southern Italian, Sicilian, Maltese, Aegean, and Italian Jewish comparators. Reference-panel sensitivity should be reported before clinal placement or proxy coefficients are converted into historical ancestry narratives.

Data Availability and Reproducibility

The manuscript is based on previously published or publicly distributed coordinate and genotype resources. The Global25 reproducibility materials include declared source and target rows, raw and grouped outputs, comparison tables, figures, screenshots, methods notes, validation scripts, and SHA-256 checksums. The AADR genotype analysis materials include stage scripts, manifests, QC summaries, matched SNP-set records, PCA/FST/ADMIXTURE result tables, figures, integrated synthesis tables, and checksums. Large raw genotype binaries are not embedded in this PDF.

The associated public repository is: https://github.com/StevenParker-collab/ashkenazi-population-structure-v1-reproducibility/tree/main/Ashkenazi_Population_Structure_Reassessed_V1_repository.

Ethics Statement

This study is a secondary analysis of previously published or publicly distributed population-genetic resources. It introduces no new biological sampling, participant recruitment, genotyping, or sequencing.

References

- Atzmon, G., Hao, L., Pe'er, I., Velez, C., Pearlman, A., Palamara, P. F., et al. (2010). Abraham's children in the genome era: major Jewish diaspora populations comprise distinct genetic clusters with shared Middle Eastern ancestry. *American Journal of Human Genetics*, 86, 850-859. <https://doi.org/10.1016/j.ajhg.2010.04.015>
- Behar, D. M., Garrigan, D., Kaplan, M. E., Mobasher, Z., Rosengarten, D., Karafet, T. M., et al. (2004). Contrasting patterns of Y chromosome variation in Ashkenazi Jewish and host non-Jewish European populations. *Human Genetics*, 114, 354-365. <https://doi.org/10.1007/s00439-003-1073-7>
- Behar, D. M., Yunusbayev, B., Metspalu, M., Metspalu, E., Rosset, S., Parik, J., et al. (2010). The genome-wide structure of the Jewish people. *Nature*, 466, 238-242. <https://doi.org/10.1038/nature09103>
- Brace, S., Diekmann, Y., Booth, T., Macleod, R., Timpson, A., Stephen, W., et al. (2022). Genomes from a medieval mass burial show Ashkenazi-associated hereditary diseases pre-date the 12th century. *Current Biology*, 32, 4350-4359.e6. <https://doi.org/10.1016/j.cub.2022.08.036>
- Bray, S. M., Mulle, J. G., Dodd, A. F., Pulver, A. E., Wooding, S., and Warren, S. T. (2010). Signatures of founder effects, admixture, and selection in the Ashkenazi Jewish population. *Proceedings of the National Academy of Sciences USA*, 107, 16222-16227.

<https://doi.org/10.1073/pnas.1004381107>

- Campbell, C. L., Palamara, P. F., Dubrovsky, M., Botigue, L. R., Fellous, M., Atzmon, G., et al. (2012). North African Jewish and non-Jewish populations form distinctive, orthogonal clusters. *Proceedings of the National Academy of Sciences USA*, 109, 13865-13870. <https://doi.org/10.1073/pnas.1204840109>
- Chazan, R. (1987). *European Jewry and the First Crusade*. University of California Press.
- Drineas, P., Tsetsos, F., Plantinga, A., Lazaridis, I., Yannaki, E., Razou, A., et al. (2019). Genetic history of the population of Crete. *Annals of Human Genetics*, 83, 373-388. <https://doi.org/10.1111/ahg.12328>
- Feldman, L. H. (1993). *Jew and Gentile in the Ancient World*. Princeton University Press.
- Grossman, A. (1995). *The Early Sages of Ashkenaz*. Magnes Press.
- Gruen, E. S. (2002). *Diaspora: Jews Amidst Greeks and Romans*. Harvard University Press.
- Hammer, M. F., Redd, A. J., Wood, E. T., Bonner, M. R., Jarjanazi, H., Karafet, T., et al. (2000). Jewish and Middle Eastern non-Jewish populations share a common pool of Y-chromosome biallelic haplotypes. *Proceedings of the National Academy of Sciences USA*, 97, 6769-6774. <https://doi.org/10.1073/pnas.100115997>
- Hammer, M. F., Behar, D. M., Karafet, T. M., Mendez, F. L., Hallmark, B., Erez, T., et al. (2009). Extended Y chromosome haplotypes resolve multiple and unique lineages of the Jewish priesthood. *Human Genetics*, 126, 707-717. <https://doi.org/10.1007/s00439-009-0727-5>
- Kopelman, N. M., Stone, L., Wang, C., Gefel, D., Feldman, M. W., Hillel, J., and Rosenberg, N. A. (2009). Genomic microsatellites identify shared Jewish ancestry intermediate between Middle Eastern and European populations. *BMC Genetics*, 10, 80. <https://doi.org/10.1186/1471-2156-10-80>
- Lawson, D. J., van Dorp, L., and Falush, D. (2018). A tutorial on how not to over-interpret STRUCTURE and ADMIXTURE bar plots. *Nature Communications*, 9, 3258. <https://doi.org/10.1038/s41467-018-05257-7>
- Lerga-Jaso, J., Novkovic, B., Unnikrishnan, D., Bamunusinghe, V., Hatorangan, M. R., Manson, C., et al. (2025). Tracing human genetic histories and natural selection with precise local ancestry inference. *Nature Communications*, 16, 4576. <https://doi.org/10.1038/s41467-025-59936-3>
- Listman, J. B., Hasin, D., Kranzler, H. R., Malison, R. T., Mutirangura, A., Sughondhabirrom, A., et al. (2010). Identification of population substructure among Jews using STR markers and dependence on reference populations included. *BMC Genetics*, 11, 48. <https://doi.org/10.1186/1471-2156-11-48>
- Mallick, S., Micco, A., Mah, M., Ringbauer, H., Lazaridis, I., Olalde, I., Patterson, N., and Reich, D. (2024). The Allen Ancient DNA Resource (AADR): a curated compendium of ancient human genomes. *Scientific Data*, 11, 182.
- Mallick, S., and Reich, D. The Allen Ancient DNA Resource (AADR): A curated compendium of ancient human genomes. *Harvard Dataverse*, V14 (v66.p1), June 8, 2026. <https://doi.org/10.7910/DVN/FFIDCW>
- Nebel, A., Filon, D., Brinkmann, B., Majumder, P. P., Faerman, M., and Oppenheim, A. (2001). The Y chromosome pool of Jews as part of the genetic landscape of the Middle East. *American Journal of Human Genetics*, 69, 1095-1112. <https://doi.org/10.1086/324070>
- Ostrer, H., and Skorecki, K. (2013). The population genetics of the Jewish people. *Human Genetics*, 132, 119-127. <https://doi.org/10.1007/s00439-012-1235-6>
- Roth, C. (1946). *History of the Jews in Italy*. Jewish Publication Society.
- Rutgers, L. V. (1995). *The Jews in Late Ancient Rome*. Brill.
- Semino, O., Magri, C., Benuzzi, G., Lin, A. A., Al-Zahery, N., Battaglia, V., et al. (2004). Origin, diffusion, and differentiation of Y-chromosome haplogroups E and J: inferences on the neolithization of Europe and later migratory events in the Mediterranean area. *American Journal of Human Genetics*, 74, 1023-1034. <https://doi.org/10.1086/386295>
- Shen, P., Lavi, T., Kivisild, T., Chou, V., Sengun, D., Gefel, D., et al. (2004). Reconstruction of patrilineages and matrilineages of Samaritans and other Israeli populations from Y-chromosome and mitochondrial DNA sequence variation. *Human Mutation*, 24, 248-260. <https://doi.org/10.1002/humu.20077>
- Tian, C., Kosoy, R., Nassir, R., Lee, A., Villoslada, P., Klareskog, L., et al. (2009). European population genetic substructure: further definition of ancestry informative markers for distinguishing among diverse European ethnic groups. *Molecular Medicine*, 15, 371-383. <https://doi.org/10.2119/molmed.2009.00094>
- Waldman, S., Backenroth, D., Harney, E., Flohr, S., Neff, N. C., Buckley, G. M., et al. (2022). Genome-wide data from medieval German Jews show that the Ashkenazi founder event pre-dated the 14th century. *Cell*, 185, 4703-4716.e16. <https://doi.org/10.1016/j.cell.2022.11.002>
- Xue, J., Lencz, T., Darvasi, A., Pe'er, I., and Carmi, S. (2017). The time and place of European admixture in Ashkenazi Jewish history. *PLOS Genetics*, 13, e1006644. <https://doi.org/10.1371/journal.pgen.1006644>
- Zoosmann-Diskin, A. (2010). The origin of Eastern European Jews revealed by autosomal, sex chromosomal and mtDNA polymorphisms. *Biology Direct*, 5, 57. <https://doi.org/10.1186/1745-6150-5-57>